

KOHLER AI Bathroom Designer & Planner

Transform bathroom constraints into an optimized,
explainable KOHLER product bundle.

AI understands the customer. Logic makes the recommendation.

UNDER THE HOOD

Understand → Optimize → Present

01 Understand
Gemini
Natural language +
image

02 Optimize
TypeScript
Hard constraints +
scoring

03 Present
react-konva
Water impact + 2D
plan

Design a bathroom that fits the user - not just a list of products.

An AI-assisted planner grounded in verified KOHLER product data, hard constraints, and explainable optimization.

THE PROBLEM

01 Too many choices

Customers must balance space, budget, style, water use, and installation constraints.

02 Fit is not obvious

A product can look right in a catalog and still be wrong for the room.

03 Sustainability is opaque

“Eco-friendly” claims are hard to compare without assumptions and numbers.

THE SOLUTION

**AI understands the customer.
Deterministic logic makes the recommendation.**

01 Understand Natural language + bathroom image
Gemini

02 Ground Verified KOHLER catalog
TypeScript + Zod

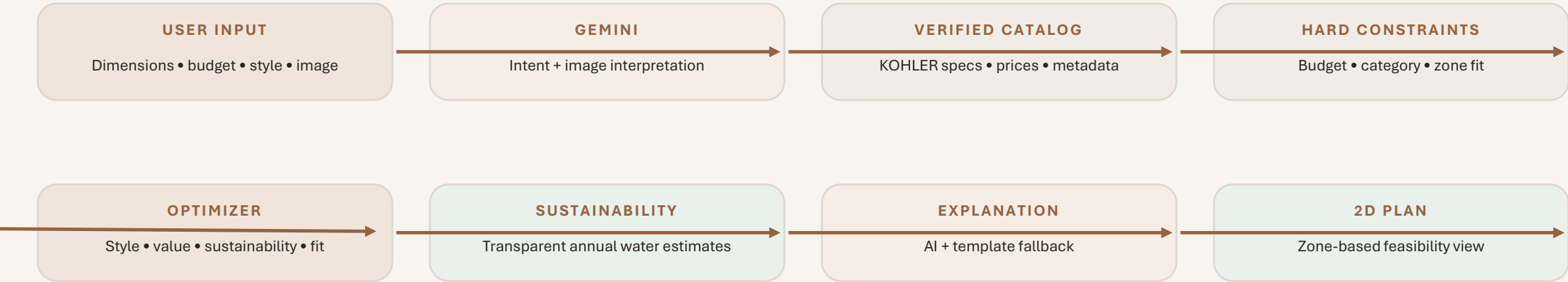
03 Optimize Hard constraints + weighted scoring
Deterministic

04 Explain Water impact + why this bundle
AI + fallback

Grounded • Explainable • Space-aware

A hybrid pipeline: AI for interpretation, deterministic engines for truth.

The LLM is deliberately bounded; critical product facts and recommendation decisions stay outside the model.



4 INNOVATIONS

- 01 Hybrid AI + deterministic**

AI understands; rules decide.
- 02 Hard / soft constraints**

Budget violations are impossible.
- 03 Transparent sustainability**

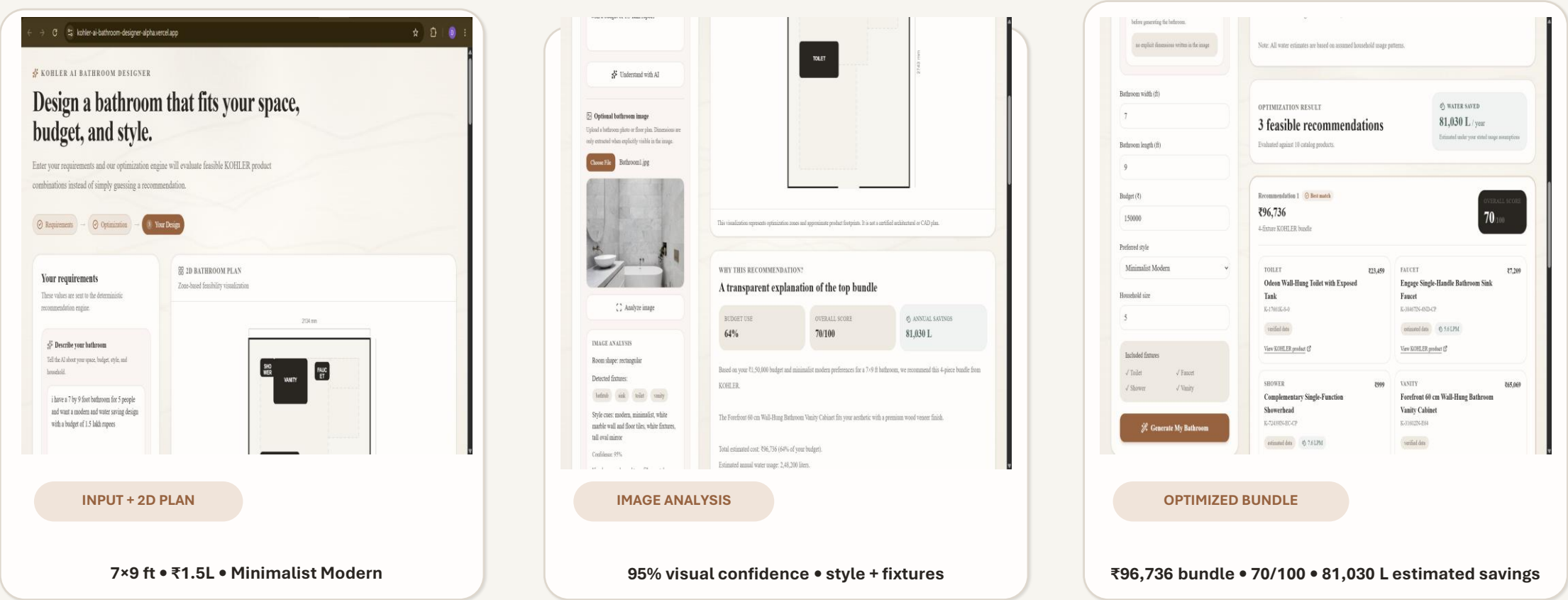
Assumptions are visible with every estimate.
- 04 Zone-based spatial fit**

Products are validated against room zones.

Tech: Next.js 15 • TypeScript • Gemini 3.5 Flash-Lite • Zod • react-konva • Tailwind/shadcn • Vercel

From a sentence and a photo to an explainable bathroom design.

Production UI: natural-language input, image analysis, ranked bundle, water impact, and 2D layout.



The product experience is designed to make the recommendation legible: what fits, what it costs, and what the estimated water impact means.

A practical decision-support layer for bathroom planning.

Designed for a real-world KOHLER-style shopping journey: fewer infeasible combinations, clearer trade-offs, transparent sustainability.

WHY IT MATTERS

-  **Space** Filters products before they reach the recommendation.
-  **Budget** Treats budget as a hard ceiling, not a preference.
-  **Sustainability** Turns water efficiency into transparent annual estimates.
-  **Trust** Shows confidence, assumptions, and source links.

BUSINESS • UX • SUSTAINABILITY

TECH STACK

Next.js 15

TypeScript

Gemini 3.5 Flash-Lite

Zod

react-konva

Tailwind + shadcn/ui

Vercel

THE PITCH

**“AI understands the customer.
Verified KOHLER data and deterministic
optimization make the recommendation.”**

Live demo: kohler-ai-bathroom-designer-alpha.vercel.app